probe an inherent advantage over unsupervised methods whose code dimensions need not correspond to individual latents. Accuracy—training a logistic probe on each method's inferred codes—provides a fairer comparison because it applies the same downstream classifier to all methods and can exploit feature combinations, not just individual dimensions. In the main text we therefore report accuracy for downstream comparisons and reserve AUC for the appendix.

MCC (end-to-end). The mean correlation coefficient (Hyvarinen and Morioka, 2016) computes the Pearson correlation between each code column and each ground-truth latent column across samples, then finds the best one-to-one matching via the Hungarian algorithm. It measures overall recovery quality: MCC = 1 when codes reproduce the true latents up to permutation and rescaling. Pearson correlation is invariant to linear scaling, so MCC does not penalise magnitude differences. However, MCC conflates dictionary quality and encoder quality—a low MCC does not tell you whether the dictionary atoms are wrong or the encoder is selecting the wrong atoms.

Dictionary diagnostics (model-level). Cosine similarity and norm ratio between matched decoder columns and ground-truth dictionary columns isolate *dictionary quality* independent of any test data or encoder. If cosine is high but MCC is low, the dictionary is good but the encoder fails. If cosine is low, the dictionary itself is the bottleneck regardless of the encoder. In our experiments, norm ratios are ≈1.0 throughout (column magnitudes are correct) while cosine similarity varies widely across SAE types (0.33–0.93), pinpointing the error as directional.

Support diagnostics (encoder-level). Support precision and recall compare the binary nonzero pattern of codes against ground truth (after Hungarian-matching via the decoder columns). These isolate *encoder quality* on the feature-selection task: does the encoder activate the right atoms, independent of magnitude accuracy? MCC cannot distinguish "activated the wrong 10 features" from "activated all 100 features and relied on magnitude differences"—the support diagnostics can. In our experiments, ReLU and JumpReLU activate ~90% of all features (precision ~0.1), revealing that they are not performing sparse selection at all.

Why all three levels are needed. Consider two failure modes that produce the same MCC:

1. An SAE with correct dictionary directions but an encoder that activates the wrong atoms. Dictionary cosine would be high; support precision would be low.
2. An SAE with wrong dictionary directions but an encoder that compensates by routing information through correlated atoms. Dictionary cosine would be low; support precision could be moderate.

MCC alone cannot distinguish these. The layered diagnostics—dictionary geometry (cosine) → feature selection (support) → overall recovery (MCC)—tell you *where* in the pipeline things break.

D.2 SCALING NUMBER OF LATENTS

Figure 14 extends the main-text Figures 6 and 8 with all six metric panels (MCC and AUC on both ID and OOD, plus accuracy). The key patterns from the main text hold across all metrics: per-sample methods maintain a small ID–OOD gap while SAEs exhibit a persistent and large gap. The frozen-decoder and refined variants consistently improve over raw SAEs, with gains most pronounced on OOD metrics.

D.3 SCALING NUMBER OF SAMPLES

Figure 15 extends Figure 10 with all six metrics. DL-FISTA benefits substantially from more data across all metrics, while SAEs plateau or degrade—confirming that the amortisation gap is not a sample-complexity issue.

D.4 VARYING SPARSITY LEVEL

Figure 16 sweeps sparsity k with $d_z = 1000$ fixed. Per-sample methods degrade gracefully as sparsity increases (the inference problem becomes harder), maintaining consistent ID–OOD performance. SAE OOD AUC converges toward chance (~0.5) at high k, further confirming the compositional generalisation failure.

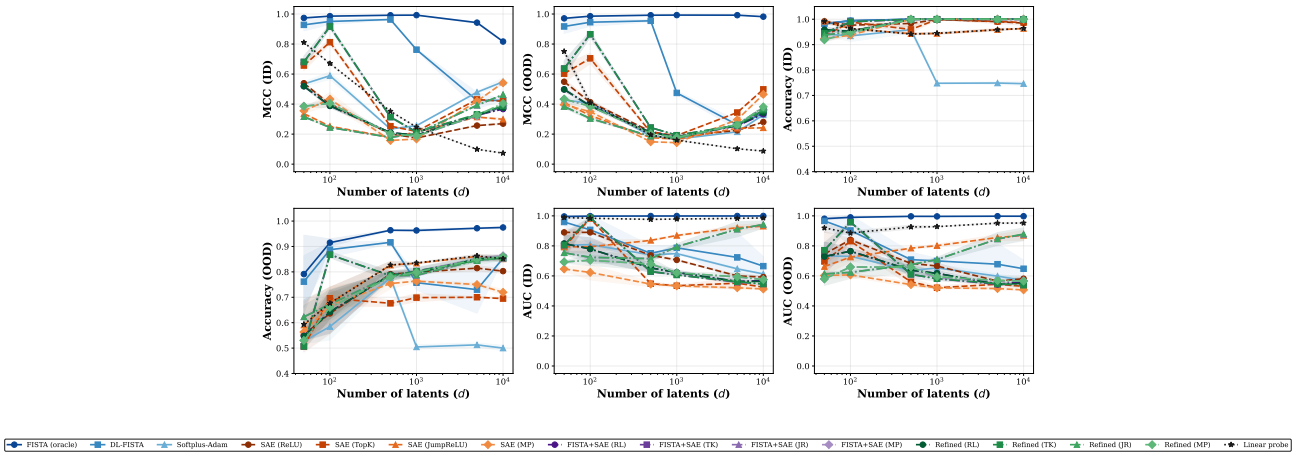

Figure 14: **All metrics vs number of latent variables.** Per-sample methods (blue) degrade uniformly across ID and OOD as d_z grows. SAEs (orange, dashed) show a persistent ID–OOD gap across all metrics. Frozen-decoder and refined hybrids (purple, green) close part of the gap by swapping inference. $k = 10$, $p = 5000$.

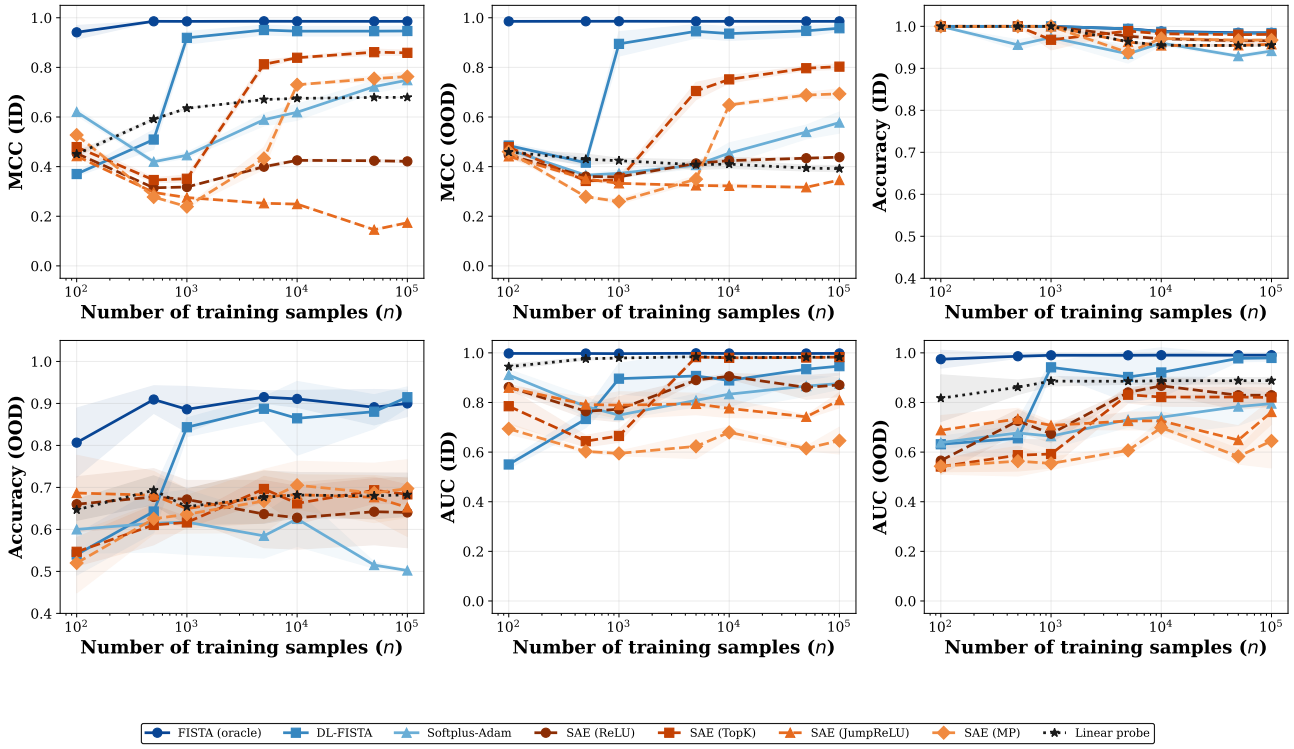

Figure 15: **All metrics vs number of training samples.** DL-FISTA's MCC and AUC improve sharply with more data and saturate by $p = 10^3$. SAE variants plateau around 0.35–0.45 MCC or degrade (JumpReLU), and OOD AUC remains scattered between 0.5–0.85 with high variance regardless of p. $d_z = 100$, $k = 10$, $d_y = 47$.

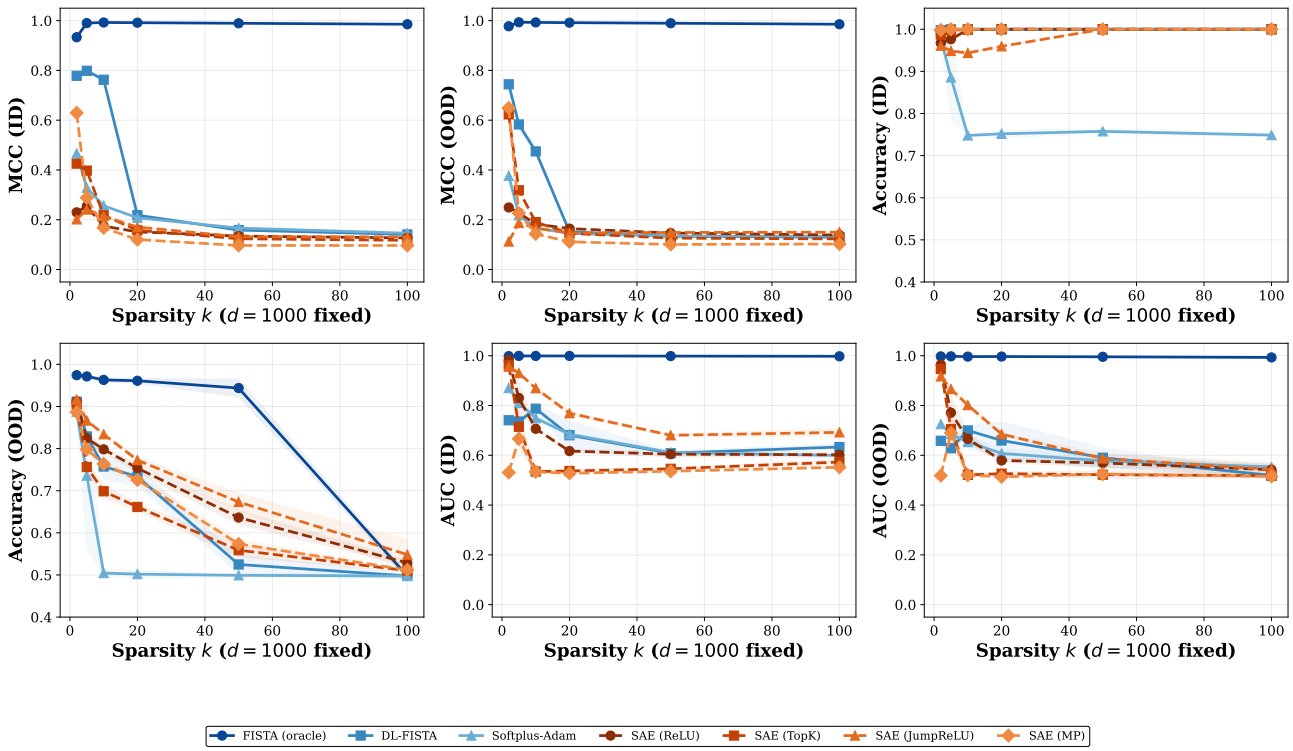

Figure 16: **All metrics vs sparsity k.** Per-sample methods degrade gracefully with increasing k; SAE OOD AUC converges toward chance at high k. $d_z = 1000$, d_y follows the CS bound.

D.5 ADDITIONAL MAIN-TEXT METRICS (V2 FIGURES)

AUC (OOD). AUC evaluates single-feature separability without a trained classifier. Note that the linear probe's AUC is computed on its supervised regression output—each output dimension targets one latent—giving it an inherent advantage over unsupervised methods whose code dimensions need not align with the label. AUC can therefore overstate the linear probe's generalisation capability relative to unsupervised methods; the accuracy metric (Figures 7 and 11) provides the fairer comparison.

D.6 CONTROLLED EXPERIMENT DETAILS

D.7 LAMBDA SENSITIVITY

The frozen decoder experiments use $\lambda = 0.1$ for FISTA, while SAE training uses $\gamma_{\text{reg}} = 10^{-4}$—a 1000× mismatch. To verify that the dictionary quality conclusion is not an artefact of this mismatch, we sweep $\lambda \in \{10^{-3}, \ldots, 2.0\}$ for both the frozen decoder and oracle conditions (Figure 30).

The oracle achieves peak MCC ~0.95 at $\lambda \approx 0.1$–0.5, confirming the correct operating regime. The frozen decoder peaks at $\lambda \approx 0.5$ for most SAE types, modestly exceeding the SAE encoder baseline for ReLU and JumpReLU (e.g., 0.4 vs 0.25 at $d_z = 5000$). However, the gap between frozen FISTA and oracle remains large at every λ, confirming that the dictionary—not the regularisation strength—is the bottleneck.

D.8 LEARNING DYNAMICS AT OTHER LATENT DIMENSIONS

D.8.1 Controlled experiments at other latent dimensions

The main text reports controlled experiments at $d_z = 100$. Below we show the same experiments at $d_z \in \{50, 500, 1000, 5000\}$ to confirm the findings hold across scales.